

Slide layouts, with notes on the optional choices

Complete Reference

Outline

- **Introduction**
 - Asymmetric layouts
- **Content Layouts**
- **Math and Academic**
- **Practical Slides**
- **Conclusions**



Introduction

01

Section open

Two ways to
mark a new
`\section`.

Option A – big navy divider (this deck's choice):

```
\AtBeginSection{\frame{\sectionpage}}
```

Option B – repeat the TOC highlighting the current section:

```
\AtBeginSection{\tocslide[currentsection]}
```

Pick one in the preamble. You can also call either macro manually inside a specific section, ignoring the `\AtBeginSection` hook.

A standard content slide

Body text uses Manrope at the same size as the original PowerPoint template.

Bullets are aligned with the Politecnico di Milano branding:

- first point of the list
- second point, with a sub-list
 - detail one
 - detail two
- third point

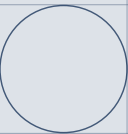
Two symmetric columns

Left column

A standard Beamer `columns` environment works as expected inside a normal frame. Use this when both sides carry equal weight.

Right column

For an asymmetric narrow-left / wide-right split (Layout 12) use `\asymtextslide` instead – see the next slides.



Asymmetric layouts

01.1

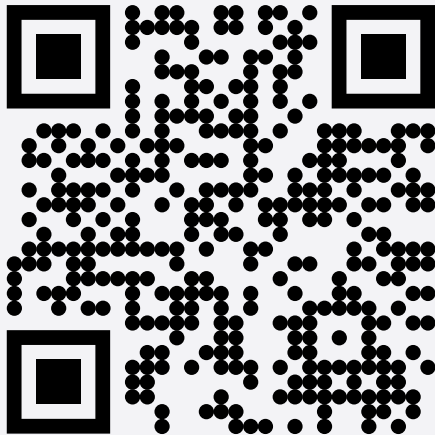
Asym. text

Short framing
remark on the
left.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur.

Asym. image

The right column
can hold an im-
age instead of
text.



Text on the left, image on the right

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

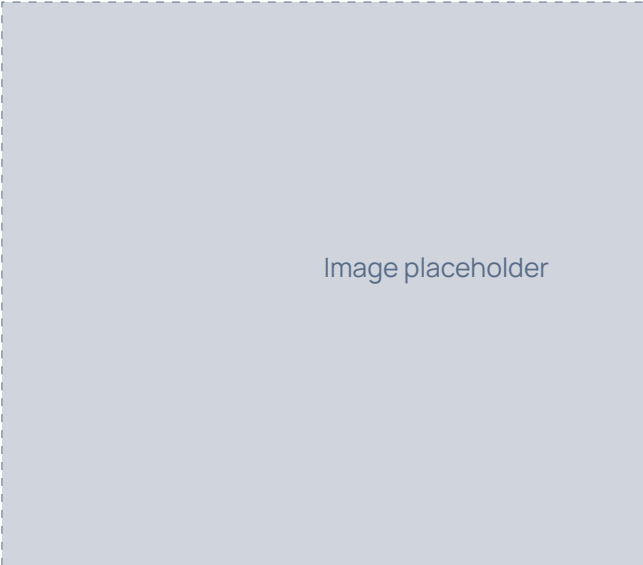


Image placeholder

Image on the left, text on the right

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

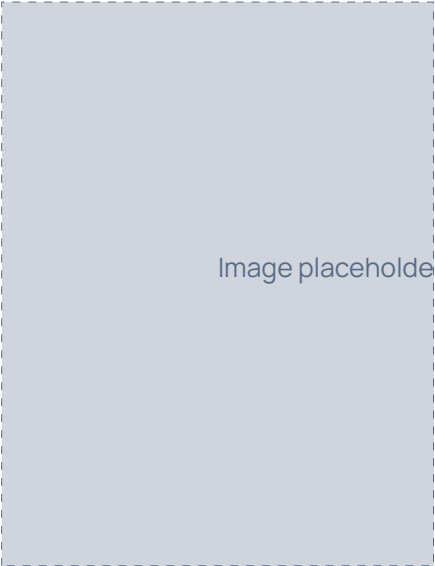


Image placeholder

Text + captioned image

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

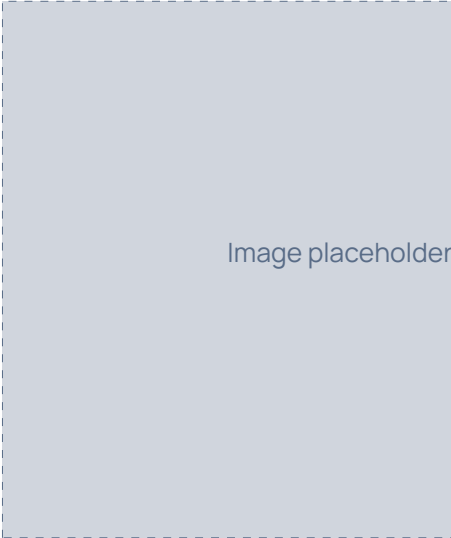


Image placeholder

Figure caption goes here

Text + two captioned images

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

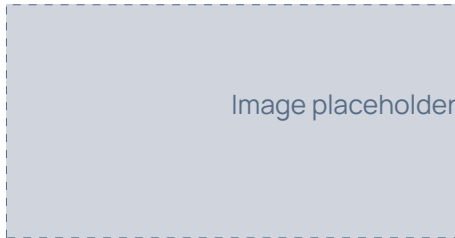


Image placeholder

First caption

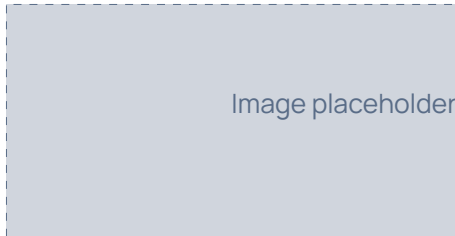


Image placeholder

Second caption



Content Layouts

02

Three symmetric columns

Column A

First approach or concept goes here.

- Advantage one
- Advantage two

Column B

Second approach for comparison.

- Advantage one
- Advantage two

Column C

Third approach closes the comparison.

- Advantage one
- Advantage two

Image placeholder

Image placeholder

Deep-dive: System Architecture

“Science is the poetry of reality.”

Richard Dawkins

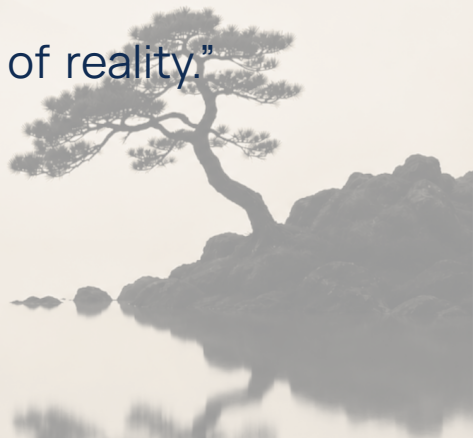
“Science is the poetry of reality.”

平常心是道



“Science is the poetry of reality.”

Richard Dawkins

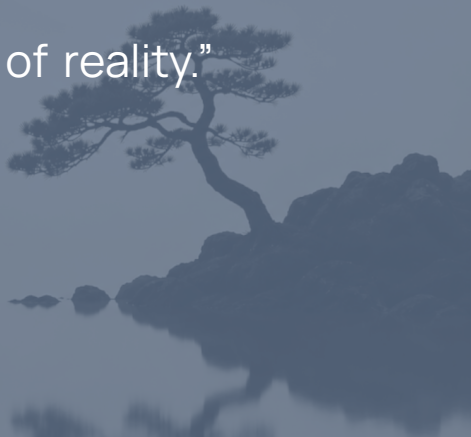


平常心是道



“Science is the poetry of reality.”

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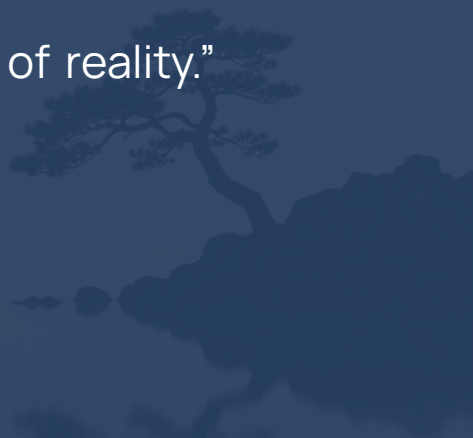
“Science is the poetry of reality.”

Richard Dawkins

平常心是道



“Science is the poetry of reality.”



Key Impact Metrics

78%

Reduction in inference latency after model distillation

123.4M

Parameters in the compressed architecture

567 mln

Total training tokens across all fine-tuning stages



Math and Academic

03

Convergence Analysis

Theorem (Banach Fixed-Point)

Let (X, d) be a non-empty complete metric space and $T : X \rightarrow X$ a contraction mapping with constant $q \in [0, 1)$. Then T has a unique fixed point $x^* \in X$, and for any $x_0 \in X$ the sequence $x_{n+1} = T(x_n)$ converges to x^* .

Convergence Analysis

Theorem (Banach Fixed-Point)

Let (X, d) be a non-empty complete metric space and $T : X \rightarrow X$ a contraction mapping with constant $q \in [0, 1)$. Then T has a unique fixed point, and the iteration $x_{n+1} = T(x_n)$ converges to it for every starting x_0 .

Use the commentary slot to state the corollary, sketch the proof idea, or comment on the implications for the talk.

Key Definition

Definition (Lipschitz Continuity)

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is Lipschitz continuous with constant $L \geq 0$ if

$$\|f(x) - f(y)\| \leq L \|x - y\| \quad \forall x, y \in \mathbb{R}^n.$$

Definition + Theorem

Definition (Lipschitz Continuity)

A function f is Lipschitz with constant L if $\|f(x) - f(y)\| \leq L \|x - y\|$ for all x, y .

Theorem (Banach Fixed-Point)

Every contraction mapping ($L < 1$) on a complete metric space has a unique fixed point, found by iterating $x_{n+1} = T(x_n)$.

Symbols test

Theorem (Symbol rendering)

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1, \quad \int_{\alpha}^{\beta} e^{-t^2} dt = \dots, \quad \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

It is possible to use the command `\polimathmanrope` in the preamble to set Manrope as the math font, which gives a more cohesive look. Otherwise, a generic sans-serif math font is used by default.

Gradient Descent

A first-order iterative optimisation algorithm for finding a local minimum of a differentiable function.

Complexity:

$\mathcal{O}(n \cdot k)$ per epoch, where n is the number of parameters and k the number of samples.

Input: $f, \nabla f, \alpha, x_0, T$

Output: approximate minimiser x^*

$t \leftarrow 0$

while $t < T$ **do**

$g \leftarrow \nabla f(x_t)$

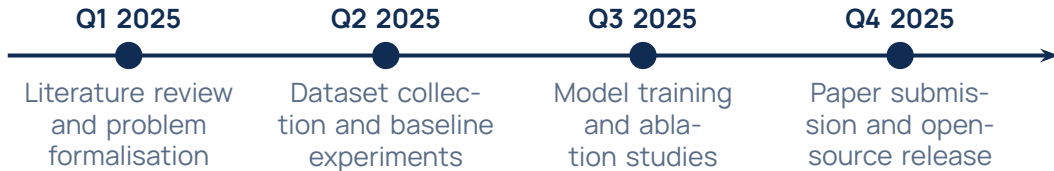
$x_{t+1} \leftarrow x_t - \alpha \cdot g$

$t \leftarrow t + 1$

end while

return x_t

Project Roadmap





Practical Slides

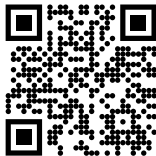
04



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Conclusions

05

Wrap-up

Final remarks before the closing slide:

- key takeaway 1
- key takeaway 2
- key takeaway 3



Thank you for your attention



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A. Author, Title of paper, Journal, 2024.



B. Other, Another work, Conf., 2023.

Contacts

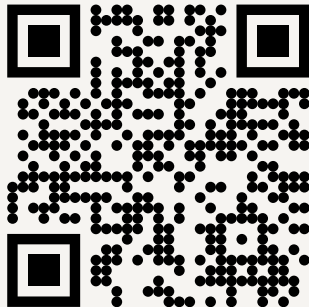
e-mail: `mario.rossi@polimi.it`



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MILANO 1863



C. Author, One more reference, Journal, 2025.

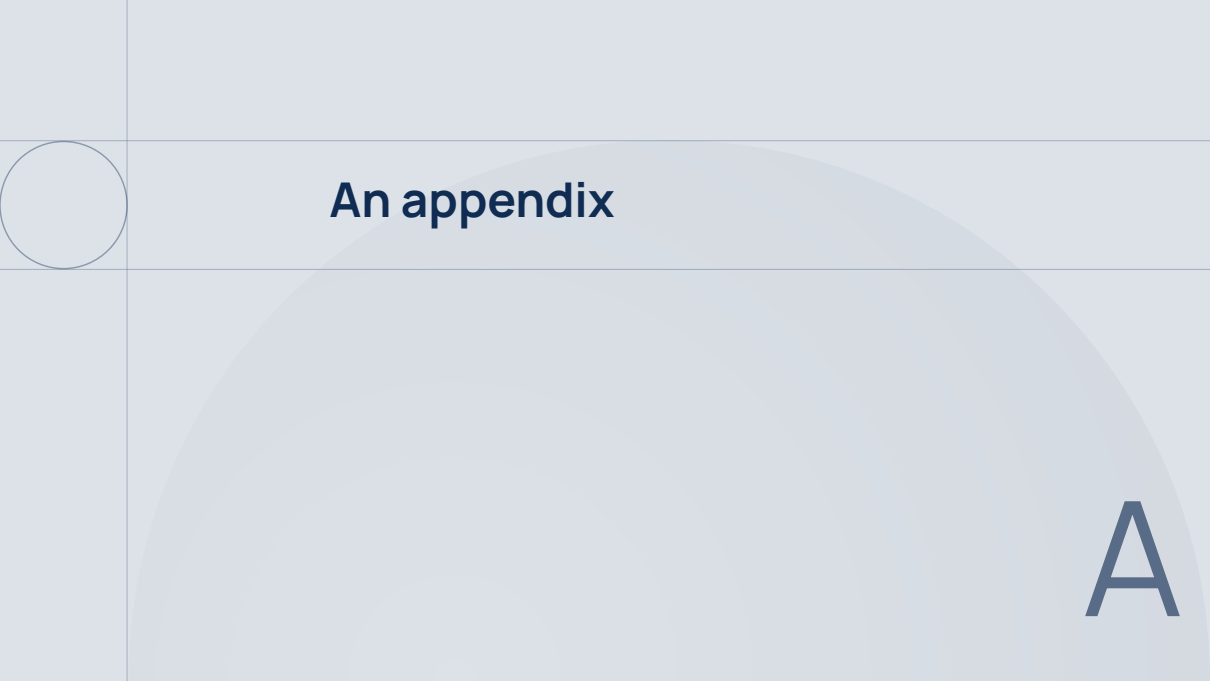


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Backup / Extra Slides



An appendix

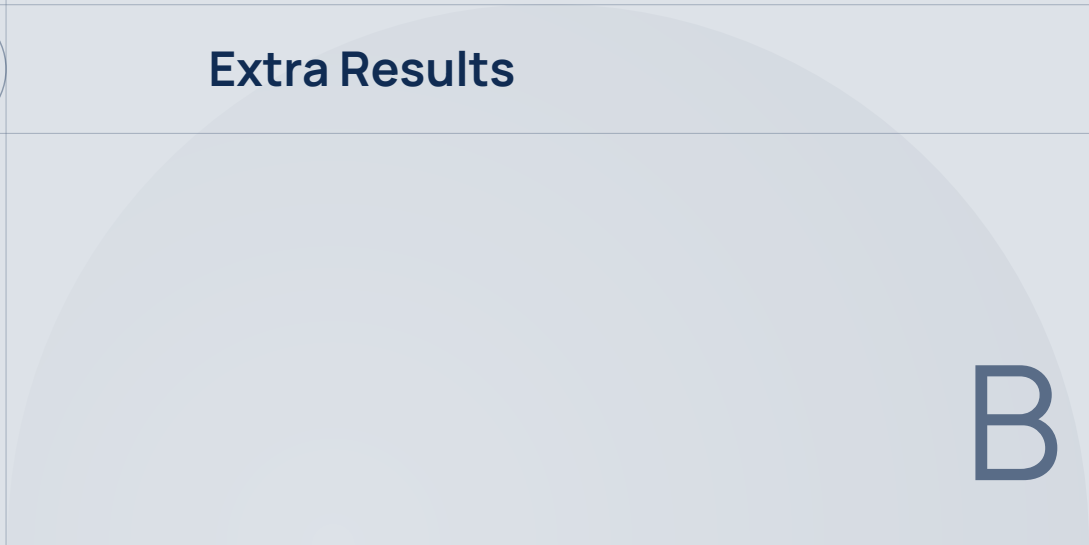
A

Supplementary: Additional Ablation

This frame is part of the backup section. It is not counted in the main slide total because `\backupmarker` was inserted beforehand and this frame uses `[noframenumbering]`.

Tip

Apply `[noframenumbering]` to every backup frame so the header bar counter stays frozen at the last main-talk slide number.



Extra Results

B

Supplementary: Extra Numbers

Another backup frame, in a second appendix block.